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Adaptive Sliding mode Control Based on RBF Neural Network Approximation for Quadrotor

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***Abstract*—This paper addresses the design of a robust adaptive sliding mode tracking control approach utilizing a Radial Basis Function Neural Network RBF NN for quadrotor. The proposed system has great advantages in dealing with nonlinearities and it has the ability to approximate uncertainties. The output of the neural network is used as a compensator parameter in order to eliminate system uncertainties. Consequently, fast error convergence in the closed loop control system can be achieved. A preliminary study to apply the system in an agricultural application using visual sensing is introduced and tested. The proposed system stability is proved by Lyapunov analysis, simulation and experimental implementation.**

***Index Terms*—Adaptive control, quadrotor, RBF neural network, sliding-mode, UAV.**

I. Introduction

The quadrotor attracts the attention of robotics researchers due to many advantages, such as maneuverability, portability, availability at low cost and the ability to fly at low speeds. As a mechanical robot, the quadrotor is a time varying uncertain dynamic system. Robustness of nonlinear controllers is affected when dealing with unknown nonlinearities, uncertainties and external disturbances. Backstepping, sliding-mode backstepping and integral backstepping are used in [1–3] as a nonlinear control to obtain stability in quadrotor. Sliding Mode Control SMC is a robust approach to deal with such unknowns, including external disturbances used in [4–6]. However, the accuracy of the mentioned controllers depends on the exact knowledge of the model and the parameters. Furthermore, the controller performance decreases if there is a variation in the parameters due to load change for example. Another control strategy is the design of adaptive control that gives more robustness to the designed controller in [7], where external disturbance and uncertainty problem is handled by an adaptive control scheme. Adaptive control method is implemented to adapt disturbance in [8] and an adaptive output feedback compensator is proposed in [9]. However, adaptive control requires a known model structure. Furthermore, SMC requires the upper bound information of model uncertainties and external disturbances. Another scheme is Fuzzy Logic Control (FLC) [10, 11]. However, the FLC lacks robustness to large variation in disturbance. Intelligent control methods like neural network have ability to provide approximation of nonlinear systems. Neural network output feedback controller is designed in [12–15]. The high performance of the Radial Base Function Neural Network (RBF NN) motivated us to utilize it in a quadrotor. The problem of high computational cost of the NN is solved by using small number of nodes. The big advantage of the proposed control system is that it doesn't require previous knowledge of the model. It works well in case of large parameters change and big imperfection of modeling. In this paper, an RBF NN is incorporated in sliding control system to strengthen the robustness of the control system. The RBF NN approximates uncertainties to improve the control capability to adapt variations in system dynamics and parameters. Such advantage makes it ideal for a load and unload application. Therefore, a preliminary study of an agricultural application "tree pollination" is introduced in this paper.

The important points of this approach are:

- *It is not required to know the upper bound of the model uncertainties and external disturbances.*
- *The utilized RBF neural network provides universal approximation to the control system to achieve high tracking accuracy.*
- *The designed controller incorporates adaptive sliding mode control and nonlinear mapping of neural network.*
- *The proposed controller can online update the unknown system dynamics.*
- *The designed controller guarantees the convergence of tracking error and it is also robust against perturbation.*

This paper is organized as follows. Section II demonstrates quadrotor dynamic model. Section III clarifies problem formulation. Section IV describes RBF neural network. Section V explains control derivation and stability analysis. Section VI briefs the proposed application. Section VII presents Matlab/Simulink simulation and Section VIII demonstrates the experimental work. The conclusion is briefed in Section IX.

II. Quadrotor Dynamics

The structure of quadrotor consists of four brushless motors and fixed rotational blades. Every rotor produces a lift force and moment. The combination of forces generated by each motor leads the quadrotor to tilt and move in different directions. The tilting around X, Y and Z axes are roll ϕ, pitch θ and yaw ψ angles respectively. The quadrotor modeling is based on Newton-Euler model. It has been demonstrated in

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a lot of research articles [16–20]. Rotor forces, moments and different frames are displayed in Figure 1.

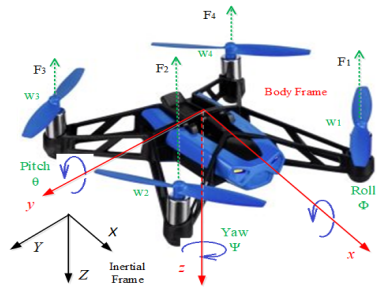

Fig. 1: Quadrotor configuration, forces and frames

The dynamic model for the translational accelerations $(\ddot{x}, \ddot{y}, \ddot{z})^T$ and the angular accelerations$(\ddot{\phi}, \ddot{\theta}, \ddot{\psi})^T$ about the body frame axis is given by force and moment analysis. It is presented in a state space form $\dot{X}{=}f(X, U)$ as follows [17, 21]:

$$\begin{aligned}
\ddot{\phi} &= \frac{i_{yy} - i_{zz}}{i_{xx}}\dot{\theta}\dot{\psi} - \frac{j_r}{i_{xx}}w_r\dot{\theta} - \frac{k_{fax}}{i_{xx}}\dot{\phi}^2 + \frac{1}{i_{xx}}u_2 \\
\ddot{\theta} &= \frac{i_{zz} - i_{xx}}{i_{yy}}\dot{\phi}\dot{\psi} + \frac{j_r}{i_{yy}}w_r\dot{\phi} - \frac{k_{fay}}{i_{yy}}\dot{\theta}^2 + \frac{1}{i_{yy}}u_3 \\
\ddot{\psi} &= \frac{i_{xx} - i_{yy}}{i_{zz}}\dot{\theta}\dot{\phi} - \frac{k_{faz}}{i_{zz}}\dot{\psi}^2 + \frac{1}{i_{zz}}u_4 \\
\ddot{x} &= -\frac{k_{ftx}}{m}\dot{x} + \frac{1}{m}u_x u_1 \\
\ddot{y} &= -\frac{k_{fty}}{m}\dot{y} + \frac{1}{m}u_y u_1 \\
\ddot{z} &= -\frac{k_{ftz}}{m}\dot{z} + g - \frac{1}{m}(\cos\phi\cos\theta)\ u_1
\end{aligned} \tag{1}$$

We denote $\ddot{X}{=}[\ddot{x}_1\ \ddot{x}_2 ...\ \ddot{x}_6]^T{=}[\ddot{\phi}\ \ddot{\theta}\ \ddot{\psi}\ \ddot{x}\ \ddot{y}\ \ddot{z}]^T$ as the state vector. The control inputs for altitude and attitude. Control inputs and motor velocities are related as follows:

$$\begin{aligned}
u_1 &= b(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \\
u_2 &= b\, l_e(\omega_1^2 + \omega_4^2 - \omega_2^2 - \omega_3^2) \\
u_3 &= b\, l_e(\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2) \\
u_4 &= c(-\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^2)
\end{aligned} \tag{2}$$

The actual input to the quadrotors are u_1, u_2, u_3 and u_4 while u_x, u_y are axillary control input Auxiliary control inputs [21] $u_x{=}(cos\phi\ \ sin\theta\ \ cos\psi\ +\ sin\phi\ \ sin\psi)$, $u_y{=}(cos\phi\ sin\theta\ sin\psi{-}sin\phi\ cos\psi)$. The axillary control input are used to calculate the desired roll $\phi_d$and desired pitch θ_d then, the roll and pitch will be controlled in u_2, u_3. The desired roll and pitch are found as [21, 22]:

$$\begin{aligned}
\phi_d &= arcsin\,(u_x sin(\psi_d) - u_y cos(\psi_d)) \\
\theta_d &= arctan(\frac{u_x cod(\psi_d) + u_y sin(\psi_d)}{cos\psi_d})
\end{aligned} \tag{3}$$

Other variables are:
ϕ, θ, ψ : roll, pitch and yaw angles respectively.
i_{xx}, i_{yy}, i_{zz} : moments of inertia about X,Y and Z axes respectively.
m : total mass.
g : gravity force.
j_r : rotor inertia.
$k_{ftax}, k_{ftay}, k_{ftaz}$:the coefficients of aerodynamic friction.
$k_{ftx}, k_{fty}, k_{ftz}$:the translation drag coefficients.
b : thrust coefficient.
d_c : drag coefficient.
l_e : length of the moment arm.
ψ_d : the desired yaw angle.
ω : motor angular velocity. (i=1,2,3,4)
ω_r : the balance around z-axis, $\omega_r = -\omega_1 + \omega_2 - \omega_3 + \omega_4$.

III. Problem Formulation

The state space vector (1) is represented as $X{=}[x_1, x_2 ... x_6]^T$ and it can be rewritten as:

$$\boldsymbol{H}\ddot{X} = F(X) + U + D(t) \tag{4}$$

The control vector is $U{=}[u_2, u_3, u_4, u_x u_1, u_y u_1, u_1]^T$. $D(t)$ is the external disturbances, $\boldsymbol{H}{=}diag(i_{xx}, i_{yy}, i_{zz}, m, m, -m)$ is the inertia matrix considering small variation of roll and pitch, and the nonlinear functions $F(\dot{X}(t))$ is given as:

$$F(\dot{X}) = \begin{bmatrix} (i_{yy} - i_{zz})\dot{\theta}\dot{\psi} - j_r\omega_r\dot{\theta} - k_{fax}\dot{\phi}^2 \\ (i_{zz} - i_{xx})\dot{\phi}\dot{\psi} + j_r\omega_r\dot{\phi} - k_{fay}\dot{\theta}^2 \\ (i_{xx} - i_{yy})\dot{\phi}\dot{\theta} - k_{faz}\dot{\psi}^2 \\ -k_{ftx}\dot{x} \\ -k_{fay}\dot{y} \\ mg - k_{faz}\dot{z} \end{bmatrix}$$

If the nonlinear function $F(\dot{X}(t))$ was exactly known, then the control would be a nonlinear control system. In real-time systems, $F(\dot{X}(t))$ and $D(t)$ are never identified accurately. Accurate estimation of unknowns and disturbances will improve system performance. The goal of this paper is to achieve the following.

Objective:

- *Provide an estimation of the unknown dynamics and disturbance to build a robust tracking control approach for the UAV quadrotor. This is to guarantee that the output tracking error convergence.*

Assumption1: The system unknowns and disturbances are continuous and bounded by a positive number.

IV. Radial Base Function Neural Network

In this section, we will address the design procedure of RBF neural network. Neural networks have the ability to approximate nonlinear continuous functions over the compact input space. They are used to adaptively learn the unknown upper bound of model uncertainties and external disturbances. Learning ability to approximate arbitrary nonlinear functions makes neural network useful tool for adaptive application.

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Its objective is to provide adaptive approximation for the uncertain and imperfect functions [23].

The RBF is a three-layer neural network, input layer, output layer and a hidden layer. Each node in the hidden layer uses a Gaussian radial basis function RBF denoted $\phi(\Gamma(t))$ as a nonlinear activation function. The hidden layer performs a nonlinear transform of the input and the output layer. It is considered as a linear combiner mapping the nonlinearity into a new space.

The algorithm of the utilized RBF neural network is [23] :

$$\begin{aligned} \varphi_i &= exp(-\frac{||\Gamma - R_i||^2}{b_i^2}) \\ Q &= \boldsymbol{W}^T \phi + \varepsilon \end{aligned} \tag{5}$$

The Gaussian function of the network is $\phi = [\varphi_1 \varphi_2 ... \varphi_n]^T$, where n is the number of neural nodes in the hidden layer, R_i is the vector of center of the ith node, b_i is used to control the smoothness of the interpolating function. The approximation error is ε. Γ is the input vector of neural network. Q is the desired estimation. The neural network weights matrix is $\boldsymbol{W}$. The network input is chosen as:

$$\Gamma = [E^T \dot{E}^T X_d^T \dot{X}_d^T \ddot{X}_d^T]^T \tag{6}$$

The error and its derivative $E = X - X_d$ and $\dot{E} = \dot{X}_d - \dot{X}$. The desired trajectory, velocity and acceleration are $X_d, \dot{X}_d, \ddot{X}_d$ respectively.

Assumption2: The approximation error ε is assumed to be bounded, $||\varepsilon|| \leqslant \varepsilon_N$ where ε_N is a positive number.

The estimated output of the neural network is

$$\widehat{Q}((x(t)) = \widehat{\boldsymbol{W}}^T \phi \tag{7}$$

With arbitrary positive initial value $\widehat{\boldsymbol{W}}$, the output of the network provides accurate estimation, as a result, the tracking error can converge to zero, as will be shown the next section.

V. Control Design

The design of the adaptive sliding mode control based on the RBF NN estimation and its stability analyses are addressed in this section. The proposed system aims to maintain asymptotic stability to the quadrotor. System block diagram is shown in Figure 2. The sliding surface is designed as:

$$S = \dot{E} + \boldsymbol{C} E \tag{8}$$

Where $\boldsymbol{C}$ is a positive definite constant matrix. The derivative of the sliding surface multiplied by the inertia matrix is given as:

$$\begin{aligned} \boldsymbol{H}\dot{S} &= \boldsymbol{H}(\ddot{X}_d - \ddot{X} + \boldsymbol{C}\dot{E}) \\ \boldsymbol{H}\dot{S} &= \boldsymbol{H}\ddot{X}_d - \boldsymbol{H}\ddot{X} + \boldsymbol{H}\boldsymbol{C}\dot{E} \end{aligned} \tag{9}$$

Substituting the system general equation (4) in (9), then:

$$\boldsymbol{H}\dot{S} = \boldsymbol{H}\ddot{X}_d - (F(\dot{X}(t)) + U(t) + D(t)) + \boldsymbol{H}\boldsymbol{C}\dot{E} \tag{10}$$

Which can be written as:

$$\boldsymbol{H}\dot{S} = Q - U \tag{11}$$

Where Q is considered as:

$$Q = \boldsymbol{H}\dot{X}_d - F(\dot{X}(t) - D(t) + \boldsymbol{H}\boldsymbol{C}\dot{E} \tag{12}$$

The control is selected as:

$$U(t) = \widehat{Q} + \boldsymbol{K}S + \varepsilon_N Sign(S) \tag{13}$$

Where $\boldsymbol{K}$ is a positive definite constant matrices. Substituting (13) in (11):

$$\begin{aligned} \boldsymbol{H}\dot{S} &= Q - \widehat{Q} - \boldsymbol{K}S - \varepsilon_N Sign(S) \\ \boldsymbol{H}\dot{S} &= \widetilde{Q} - \boldsymbol{K}S - \varepsilon_N Sign(S) \end{aligned} \tag{14}$$

Where $\widetilde{Q} = Q - \widehat{Q}$ is the estimation error in the dynamic system. The following Lyapunov function is used to analyze the stability of the system [23]:

$$V = \frac{1}{2} S^T \boldsymbol{H} S + \frac{1}{2} tr(\widetilde{\boldsymbol{W}}^T \boldsymbol{B}^{-1} \widetilde{\boldsymbol{W}}) \tag{15}$$

$$\dot{V} = S^T \boldsymbol{H}\dot{S} + tr(\widetilde{\boldsymbol{W}}^T \boldsymbol{B}^{-1} \dot{\widetilde{\boldsymbol{W}}}) \tag{16}$$

Where $tr(.)$ is the trace of a matrix, $\boldsymbol{B}$ is the adaptation gains positive definite matrix. By substituting (14) in (16), it can be found that:

$$\dot{V} = S^T(\widetilde{Q} - \boldsymbol{K}S - \varepsilon_N Sign(S) + \ tr(\widetilde{\boldsymbol{W}}^T \boldsymbol{B}^{-1} \dot{\widetilde{\boldsymbol{W}}}) \tag{17}$$

Using (5) and (7), then:

$$\begin{aligned} \dot{V} &= S^T(\widetilde{\boldsymbol{W}}^T \phi + \varepsilon - \boldsymbol{K}S - \varepsilon_N Sign(S) + tr(\widetilde{\boldsymbol{W}}^T \boldsymbol{B}^{-1} \dot{\widetilde{\boldsymbol{W}}}) \\ \dot{V} &= S^T(\varepsilon - \boldsymbol{K}S - \varepsilon_N Sign(S) + tr(\widetilde{\boldsymbol{W}}^T (\boldsymbol{B}^{-1} \dot{\widetilde{\boldsymbol{W}}} + \phi S^T)) \end{aligned} \tag{18}$$

By letting $(\boldsymbol{B}^{-1}\dot{\widetilde{\boldsymbol{W}}} + \phi S^T) = 0$, we can find $\dot{\widetilde{\boldsymbol{W}}} = -\boldsymbol{B}\phi S^T$ and for small variation of $\boldsymbol{W}$ we can consider the following adaption law:

$$\dot{\widehat{\boldsymbol{W}}} = -\boldsymbol{B}\phi S^T \tag{19}$$

Then we have:

$$\begin{aligned} \dot{V} &= S^T(\varepsilon - \boldsymbol{K}S - \varepsilon_N Sign(S)) \\ \dot{V} &= \sum_{i=1}^{n} s_i(\varepsilon_i - k_i s_i - \varepsilon_N |s_i|) \end{aligned} \tag{20}$$

Based on assumption2, we have $s_i(\varepsilon_i - k_i s_i - \varepsilon_N |s_i|) \leqslant 0$ as a result $\dot{V} \leqslant 0$.

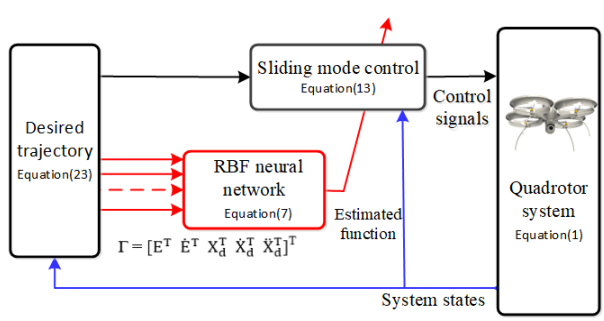

Fig. 2: System block diagram

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VI. Application: Tree Pollination

In some applications, the quadrotor is required to approach tree crowns, this could be for picturing, filming, spraying pesticides or pollinating. In this application, the quadrotor mission is to reach a tree crown in order to spray pollinating materials. It is assumed that GPS position of the desired tree is given and stored in the quadrotor. Approaching tree crown could be achieved easily if the GPS error margin is zero, but this not possible. Image processing algorithm is used to detect the tree and to provide accurate position.

In this application the goal is to obtain the pose from vision rather than perform a complex and heavy navigation.

Camera Vision System:

A pinhole camera model used in the quadrotor as in Figure 3. We assume quadrotor frame and camera frame are identical. Under a pinhole camera model, the coordinates of the target point can be given as [24]:

$$x_b = v_i \frac{z_b}{f} \qquad y_b = u_i \frac{z_b}{f} \tag{21}$$

The coordinates of the target point with respect to the quadrotor body frame is $P_b = (x_b, y_b, z_b)$, f is the camera focal length, (v_i, u_i) is the coordinates of tree crown centroid in the image plan. If f and z_b are known, the quadrotor will be able to detect $x - y$ distance to the object. Tree pose in inertial frame $P_I = P_b + Q_I$, where Q_I is quadrotor pose in the inertial frame. In practical implementations, the quadrotor is tilted by roll and pitch angles, a correction must be made to (v_i, u_i) as detailed in [24].

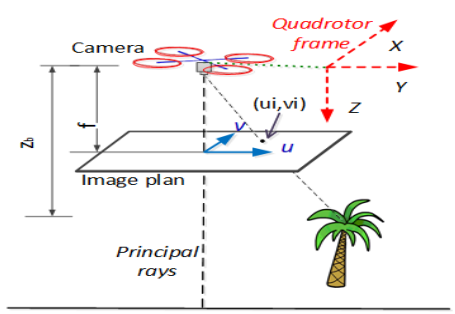

Fig. 3: Image plan projection of the feature point

VII. Simulation

In this section, the simulation results are presented to validate the effectiveness of the controller. Simulation is carried out by Matlab/Simulink software for the studied quadrotor described in equations (1). The parameters are used based on parrot-rolling spider quadrotor [25] as given in Table 1.

m	=0.068 [kg]
i_{xx}	=0.686e-3 [kg.m2]
i_{yy}	=0.092e-3 [kg.m2]
i_{zz}	=0.1366e-3 [kg.m2]
g	9.81 [m/s2]
j_r	= 1.0209e-7 [kg.m2]

TABLE 1: Quadrotor parameters

In the simulation we adapted control law in (13) which used the RBF algorithm in (5) and (7) in order to provide $\widehat{Q}(X(t)$ with the adaptive law in (19). The following reference trajectory is built to assess the quadrotors tracking performance:

$$\begin{aligned} x_d &= \begin{cases} 0 & for\ t < 3 \\ 0.5sin(0.5t) & for\ t \geq 3 \end{cases} \\ y_d &= \begin{cases} 0 & for\ t < 3 \\ 0.5 + 0.5sin(0.5t) & for\ t \geq 3 \end{cases} \\ z_d &= 1 \quad \psi_d = 0 \end{aligned} \tag{22}$$

The desired and the obtained trajectory in 3D space are shown in Figure 4. The trajectory in translation and orientation movement is shown in Figure 5 and Figure 6. Both figures 5 and 6 show good tracking of the desired trajectory.

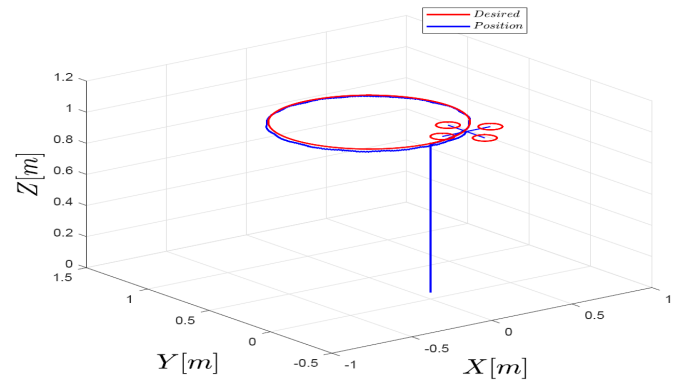

Fig. 4: Trajectory in 3D-space

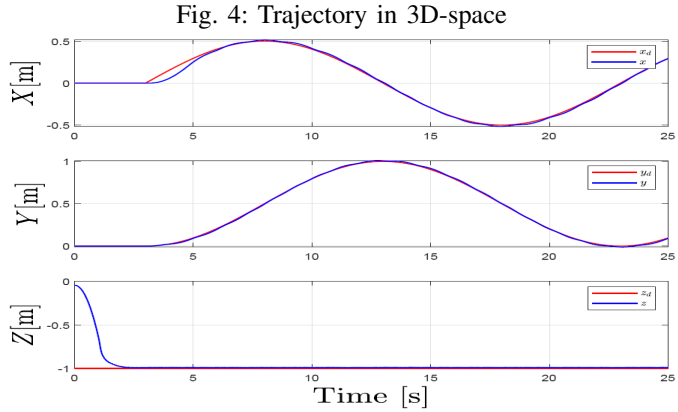

Fig. 5: Position and altitude trajectory

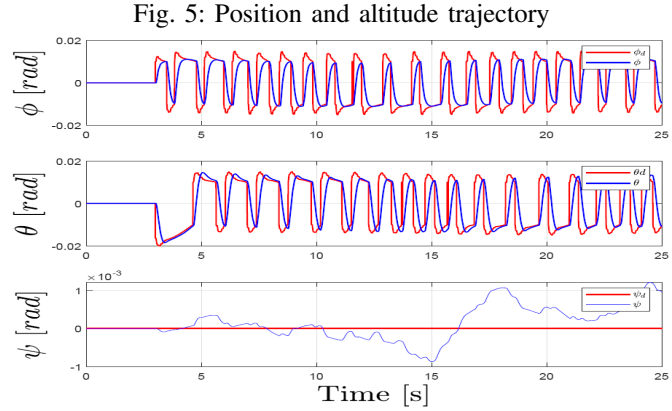

Fig. 6: Euler angles response

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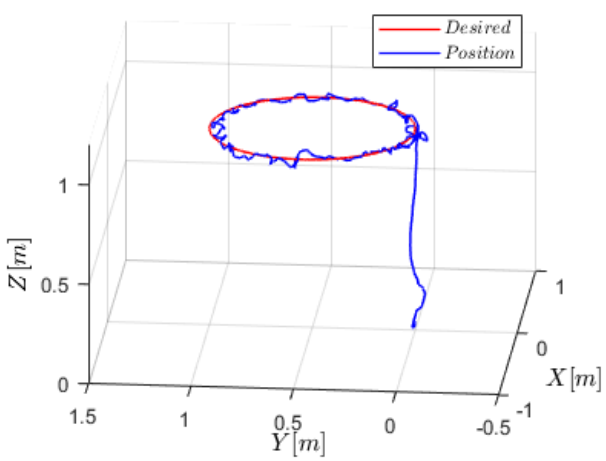
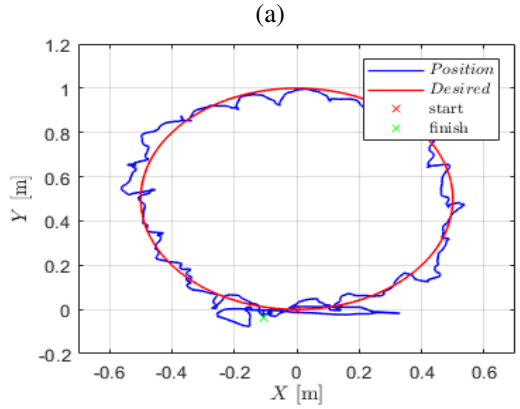
VIII. Experimental results

A. RBF NN Testing

The performance of the proposed system is demonstrated experimentally in this part. The practical implementation is based on Simulink support package for PARROT mini drone. Parrot has an integrated IMU with a three-axis gyroscope three-axis accelerometer, a compass, as well as an altitude sonar and pressure sensors. It is also equipped with a battery that retained autonomy up to 8 minutes. More information about the setup can be found in [25]. The reference trajectory is described as in the simulation part. Experimental results are presented in Figure 7-15. Figure 7-a shows the 3D task space tracking of the desired trajectory, while Figure7-b shows a top view of the tracking of the trajectory in $x-y$ direction. Figure 8 shows trajectory tracking for each axis. Both figures 7 and 8 show good tracking during the whole operation time. Orientation angles response is displayed in Figure 9 which shows fast response of the angles to stabilize the system. Figure 10 and 11 show error signals, small value of the errors can be noticed, the maximum error is 10 cm. The RMS value of the translational errors are $e_x = 0.0565, e_y = 0.0369, e_z = 0.1862$. It can be seen in Figure 12 that the chattering is small. The proposed controller shows good tracking of the desired trajectory with accuracy.



(a)



(b)

Fig. 7: (a) The trajectory in 3D (b) The trajectory in x-y

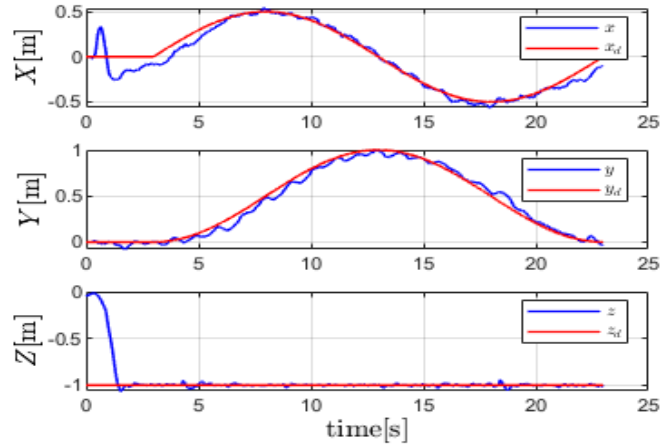


Fig. 8: Position and altitude trajectory

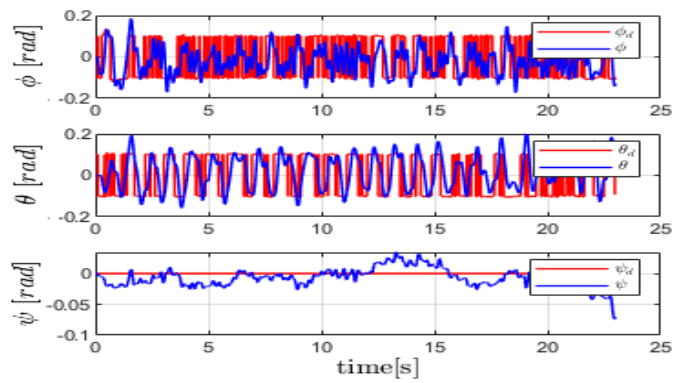


Fig. 9: Euler angles response

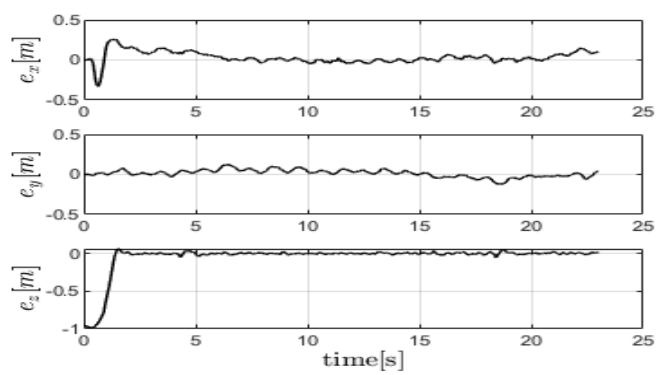


Fig. 10: Errors in position and altitude

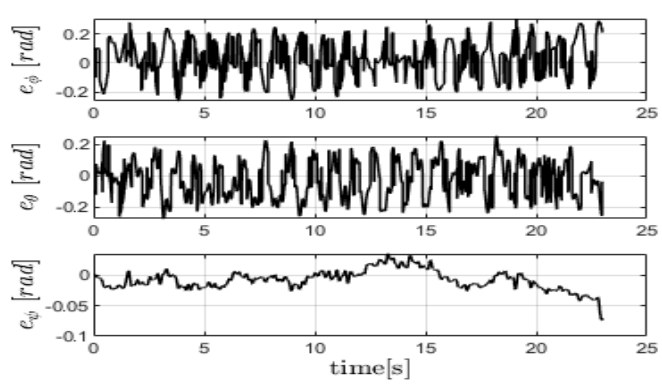


Fig. 11: Errors in orientation

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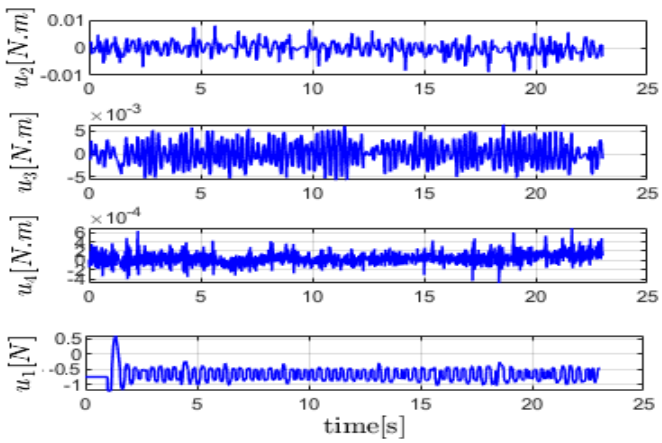

Fig. 12: Control signals

B. Tree Pollination Implementation

Implementation part is based on Parrot "Rolling Spider" quadrotor. It is equipped with a sonar and pressure sensors to measure altitude, an accelerometer, a gyroscope and a vertical camera (120X160) pixels.
The goal is to obtain the pose from vision rather than design a navigation system.
In the beginning, the quadrotor flies close to the tree as shown in Figure 13-a. Image processing is performed to detect tree position accurately, then it moves to hover over the tree, Figure 13-b. Image processing algorithm is created and tested on the quadrotor, it is built to detect tree crown centroid as a feature point in the image.

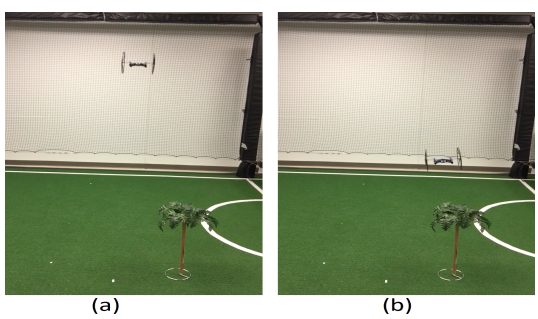

Fig. 13: Quadrotor approaching the tree

The main idea in tree detection algorithm is to detect the edges of the leaves in the green level of the image. As seen in Figure 14, image processing algorithm works in four steps. A picture of the tree is taken when the quadrotor reaches the GPS position (based on the assumption), Figure 14-a. In the beginning, we detect the edges of the leaves in the green level of the image, this is to reduce computational cost of image processing Figure 14-b. The used edge detector is "Sobel" type [26]. Then, "closing" operation is performed to make detected edges from one object, Figure 14-c. After that, "erosion" operation is processed to remove small anomalies. In the end, the tree crown is formed, Figure 14-d, and the tree centroid is determined.

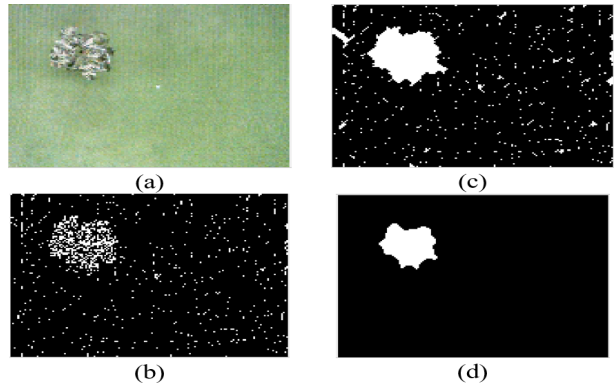

Fig. 14: Image processing algorithm

The feature point coordinates $(v_i, u_i) = (17, -38)$ pixels and the height is found by a sonar sensor $|z_b| = 1.2cm$. the focal length of the camera $f = 150$ pixels. It is found that $(x_b, y_b) = (13.6, -30.4)cm$. Quadrotor trajectory is shown in Figure 15. In stage1 the quadrotor moves to point $(1, 0, -1.8)$, in stage2, it analyzes the image to detect the tree crown position, in stage3 it approaches the crown, finally in stage4, the quadrotor is hovering over the tree.

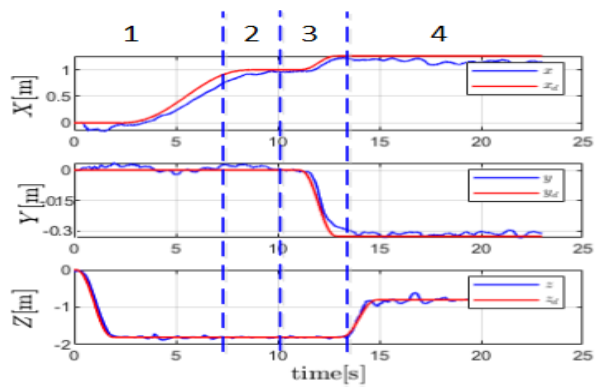

Fig. 15: Quadrotor trajectory

IX. Conclusion

A robust adaptive sliding mode tracking controller based on RBF NN approximation is accomplished in this paper. A prior knowledge of the dynamic model is not required in this scheme. By adaptively estimating uncertain functions using the RBF neural network, the effects of system uncertainties are reduced and the controller obtained asymptotic error convergence. A preliminary study to apply the system in an agricultural application using visual sensing is introduced and tested. The stability of the whole system is studied and proved by Lyapunov analysis. Simulation and experimental work are developed to test the performance of the system which verified the effectiveness of the controller.

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